Lab 6 – Branch Coverage Analysis Report

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Application: `password_policy.py`
Test file: `BranchCoverageTest_AM039_application.py`

Commands used

```
python -m coverage run --branch -m pytest BranchCoverageTest_AM039_application.py
python -m coverage report -m
```

Result

	Starter tests	After new tests
Tests	2	36 (all passed)
Branches in `password_policy.py`	36	36
Partial branches (BrPart)	9	0
Branch coverage	**52%**	**100%**

Screenshot



Analysis

- Branch coverage checks that every decision (`if` , `elif` , `for`) is tested for both its **True** and its **False** outcome.
- Tests were added so that each decision goes both ways:
  - **Length limits:** 7 vs 8 characters, and 64 vs 65 characters
  - **Common password:** a common password vs a normal one, including the case-insensitive `"PassWord"`
  - **Missing character types:** a password missing each type (uppercase, lowercase, digit, special) vs one that has it
  - **Empty password `""` :** the character loop does not run at all
  - **Strength:** 13 vs 14 characters (good vs strong), and 2 vs 3 issues (weak vs very weak)
  - **`suggest_fix` :** every kind of first issue, including `"too long"` , which reaches the final `return`
- All 36 branches of `password_policy.py` are now covered, so branch coverage is **100%** with 0 partial branches, above the 90% target.
- Branch coverage is stronger than code coverage. It forced tests on the “False” side of each condition and on the boundary values, which line coverage does not require.